Fundamentals - Day 1

Modeling & Design Patterns (Target 90 minutes) Audience – Data Engineers, Data Analysts

- Inmon Vs Kimbal Vs Data Vault
- Realtime Vs Batch
- Conceptual / Logical / Physical

Agile Dimensional Modeling (BEAM) (Target 90 minutes) Audience – Data Engineers, Data Analysts

- Business Event Analysis and Modeling (BEAM✳): an agile approach to dimensional modeling
- Thinking dimensional using the 7Ws (who, what, when, where, how many, why & how)
- Data stories, themes and BEAM✳ tables: modeling BI data requirements by

Intro to Facts & Dimensions (Target 120 minutes) Audience – Data Engineers, Data Analysts

- Facts and Dimensions
- Grouping Dimensions and Facts
- Keys and History
- Using a Star Schema
- Querying Facts
- Browsing Dimensions

Intro to Star Schema (Target 120 minutes) Audience – Data Engineers, Data Analysts

- Dimension Table Features
  - Surrogate Keys and Natural Keys
  - Rich Set of Dimensions
  - Grouping Dimensions into Dimension Tables

- Fact Table Features
  - Fact Tables and Processes
  - Capturing Facts
  - Grain
  - Sparsity
  - Degenerate Dimensions

- Slowly Changing Dimensions
    - Type 1 Change
    - Type 2 Change

Deep Dive - Day 2

Designing Facts (Target 120 minutes) Audience – Data Engineers)

- Fact types: transactions, periodic snapshots, accumulating snapshots
- Fact additivity: additive, semi-additive and non-additive measures
- Fact performance and usability: indexing, partitioning, aggregating and consolidating facts
- Fact less facts

Designing Dimensions (Target 120 minutes) Audience – Data Engineers)

- Conformed Dimensions
- Dimensions and Drilling Across (Hierarchies)
- Grouping Dimensions into Tables
- Breaking Up Large Dimensions
- Dimension Roles and Aliasing
- Avoiding the NULL
- Behavioral Dimensions
- Modeling time as dimensions and facts

Hierarchies and Snowflakes (Target 60 minutes) Audience – Data Engineers

- Drilling
    - The Concept of Drilling
    - The Reality of Drilling
- Attribute Hierarchies and Drilling
    - The Attribute Hierarchy
    - Drilling Within an Attribute Hierarchy

    - Other Ways to Drill
    - Documenting Attribute Hierarchies
- Snowflakes
- Avoiding the Snowflake
- Embracing the Snowflake
- Outriggers

Performance (Target 60 minutes) Audience – Data Engineers)

- Aggregated Tables.
    - How to load aggregates
    - Using Aggregates
- Derived Schemas - Restructuring Dimensional Data
- Pivoted Tables
- Merged Fact Tables
- Sliced Fact Tables
- Semantic layer design with Materialization.

Design ETL Processes : (Target 60 minutes) Audience – Data Engineers)

- How to load Facts & Dimension.
- Load Optimization
- ETL design to support SCD/EDW/Aggregation etc

Example EDW case studies : **Star schema & BI Consumption Patterns – 2 Hrs**